For this project, I chose to analyze and visualize the step count data for some of my friends I've made here at Northwestern in my first quarter. This typically is a time of adjusting to the uniqueness of the college experience, so I wondered how physical activity might or might not change. The data was taken from the iPhone health app, which implicitly tracks steps on a daily basis, specifically for five students: Adela, Hulan, Isabella, Julia, and Lisa. The data captures daily step counts throughout this first quarter, specifically ranging from September 9th, move-in day, to December 12th, the end of our first finals week.

The raw data came in the form of an XML file. So, using R's xml2 and tidyverse packages, I successfully extracted the data I wanted. I learned that the way our phones store our step counts are in little periods of time. So, throughout a single day, there were several entries of steps throughout the day. As a result, I had to group them by date, and summarize and sum the little snapshots in a single day, to get the total step count for that day. Additionally, step count was not the only data that was present in the Health app's exported data. For example, Julia owns an Apple Watch, which had additional statistics tracked, such as sleep or heart rate. I had to filter out those so only the steps counted by her iPhone were tracked, to be consistent with the rest.

I also augmented the raw step data with some information manually. I thought that including major (school) information and dorm location might be interesting to see with the step data. From what I've noticed so far, there seems to be a North Campus/South Campus divide where the South is generally perceived as quieter, which I thought might translate to fewer step counts. However, after plotting all the data, I did not notice any significant differences between Lisa, the only one I got data for that lived in the North, and the rest of them.

For creating the plot itself, I used a faceted time series plot, choosing to facet by student, as not to over clutter the plot with five squiggly messes of lines. Additionally, to make the graph more readable, I used the smooth geom to create a smoothed trend line that I layered on top of the raw step data to provide the viewer with a quick glimpse of the trend. For example, you can clearly see the dip across the board that occurred over Thanksgiving break. Further, I chose to add dashed lines vertically to mark key events over the course of the quarter. The ones I chose as most important were Wildcat Welcome (orientation for first-year students), First Day of Classes, Thanksgiving Break, and Finals Week. I debated adding a note for midterms, however, the fluctuations and differing times of when each person had their midterms in their classes, were too varied to add anything meaningful.

For colors/styling, I googled a theme package that allowed me to remove the background boxes and messy gridlines. Using the minimalism theme, I removed everything that I deemed to clutter the plot. I asked each student what color they would want to be, and used colors accordingly. Lisa chose purple, so I made her Northwestern Purple, to add a subtle connection to the school. Instead of adding a legend, I chose to superimpose the name of the student, their school, and dorm, only their graph itself using a text geom. This way, you could know that information immediately, rather than having to find it someplace else.

Overall, I think this plot successfully narrates the physical story of our first quarter at Northwestern. For broader trends, there is a peak at Wildcat Welcome, a slight dip for most in October, presumably because of Midterms, and the drastic dip for Thanksgiving. It was also interesting to see how some of them during finals week utilized that time to explore the city, hence their steps count skyrocketed during that time. However, for others, specifically Isabella, that time seemed to be spent stagnic, presumably in the library studying.

As mentioned earlier, I didn't notice a huge difference in dorm location. But, a theory I have is that many of the individuals who lived south, had majors such as McCormick or ISP (integrated sciences program), which forced them to walk North everyday for class. On the other hand, Isabella lives on South Campus and has most classes there as well.

I would have liked to plot data as well, but unfortunately I purchased a new phone halfway through the quarter, and none of my data transferred over.